



## Research

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### Author for correspondence:

Cai-Hua Xiong  
e-mail: [chxiong@hust.edu.cn](mailto:chxiong@hust.edu.cn)

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# Assessing the manipulative potentials of monkeys, apes and humans from hand proportions: implications for hand evolution

Ming-Jin Liu, Cai-Hua Xiong and Di Hu

Institute of Rehabilitation and Medical Robotics, State Key Lab of Digital Manufacturing Equipment and Technology, Huazhong University of Science and Technology, Wuhan, Hubei 430074, People's Republic of China

**id** M-JL, 0000-0002-9722-3870; C-HX, 0000-0003-2326-0289; DH, 0000-0002-0582-3108

The hand structure possesses a greater potential for performing manipulative skills than is typically observed, whether in humans or non-human anthropoids. However, a precise assessment of the potential manipulative skills of hands has been challenging, which hampers our understanding of the evolution of manipulative abilities in anthropoid hands. Here, we establish a functional model to quantitatively infer the manipulative potentials of anthropoid hands based on hand proportions. Our results reveal a large disparity of manipulative potentials among anthropoid hands. From the aspect of hand proportions, the human hand has the best manipulative potential among anthropoids. However, the species with a manipulative potential closer to that of humans are not our nearest relatives, chimpanzees, but rather, are certain monkey species. In combination with the phylogenetically informed morphometric analyses, our results suggest that the morphological changes of non-human anthropoid hands did not coevolve with the brain to facilitate the manipulative ability during the evolutionary process, although the manipulative ability is a survival skill. The changes in non-human anthropoid hands may have more likely evolved under selective pressure for locomotion than manipulation.

## 1. Introduction

The ability to perform various and skillful manipulations using the hand is one of the major characteristics of anthropoids (e.g. [1–4]) and allows these organisms to exploit the resources in their environment. Anthropoids use their hands not only for processing food items but also for manipulating non-edible objects during daily life [5]. Object manipulation is an important survival skill and has been assumed to be a precursor of tool use [6]. Understanding the origin and evolution of manipulative abilities in human and non-human anthropoids is a fundamental question in evolution research. Manipulative abilities are traditionally considered to have continuously evolved among anthropoids and are most pronounced in humans, who have larger brains and enhanced intelligence to improve the ability [7,8]. Undergoing a long evolutionary process, anthropoid hands have also undergone notable changes and formed specific structures. Thus, the question arises as to whether the evolutionary changes of hand structures help to facilitate the manipulation. To answer the question, it is necessary to elucidate which hand structures are suited for manipulation. The hand is generally considered to be skilled at manipulation in species that can perform more complex manipulative tasks. However, the hand structure itself possesses a greater potential for manipulation than is typically realized, whether in humans or non-human anthropoids. For instance, after long-term intensive training, humans can further exploit the potential skills and become experts in areas that require outstanding manipulative capabilities, such as professional athletes or musicians [9]. Captive capuchin monkeys can further improve their manipulative abilities and become

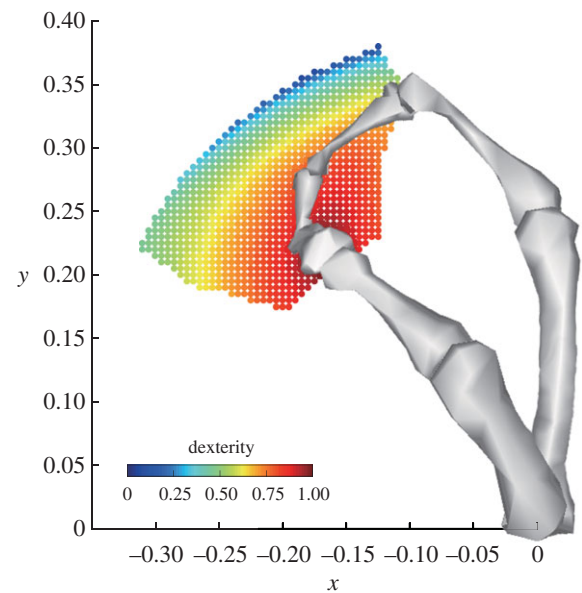
more skilled in tool use in both frequency and diversity [10]. The manipulative potential given by hand structure itself has not been fully developed, due to the limits of many factors such as environment. Thus, whether the hand structure of an anthropoid species is suitable for manipulation should focus on the manipulative potential given by the hand structure itself. Determining methods of inferring potential manipulative skills is becoming an urgent and challenging problem. Effective inferences of the potential manipulative capacities of hand structures resulting from different evolutionary changes may also be used to determine the factors that have provided humans with the most advanced manual manipulative skills and could serve as inspiration for the design of robotic hands [11].

Many researchers have tried to infer the manipulative potential of hand structure from a lot of morphological features [12,13]. A most obvious and critical morphological feature is the length ratio between the thumb and fingers. The hand with a long thumb is considered advantageous for manipulation. However, the predicted way is unable to quantitatively infer the manipulative potential from hand proportions and may be not reliable. The prediction is based on the comparative studies among different species, but lacks a direct functional link between the morphology of hand proportions and manipulation. To build the functional link and quantitatively assess the manipulative potential of different anthropoid species from hand proportions, this study relied on the work from a previously published study [14] as well as information from robotics to develop a functional model. A complex task of in-hand manipulation using the tips of the thumb and forefinger (figure 1) is used to estimate the manipulative potential in our model (see the electronic supplementary material, Note S1 for an explanation of the choice). We first applied the functional model to infer the manipulative potentials of the hands in extant anthropoid samples, including New World monkeys, Old World monkeys, apes, and humans. Manipulative potential is measured by a global manipulation index (GMI), which is based on the reachable workspace index and dexterity index in robotics. Second, we studied the major variations of hand-length proportions among the anthropoid hands and assessed the factors that facilitated the manipulation. Third, we performed evolutionary analyses of hand proportions and inferred the evolutionary history of anthropoid hands and the corresponding manipulative potentials over time. Fourth, we studied the effect of the constraints of joint mobility on manipulation by adding the actual range of motion of joints across different anthropoid species to the assessment of the manipulative potential. By means of this analysis, we hope our study can not only help to understand the effect of hand proportions on manipulation, but also give a deeper understanding of hand evolution.

## 2. Material and methods

### (a) Samples

Our analysis is based on a dataset that includes a total of 137 hand samples from 13 anthropoid species, including humans, apes, Old World monkeys, and New World monkeys (electronic supplementary material, table S1, the raw data are provided in [14]). Note that we concentrate on anthropoids and do not include all primates in the comparative analyses because most prosimians present hands with different morphologies and



**Figure 1.** Functional model of in-hand manipulation between the tips of the thumb and forefinger. The thumb tip touches the tip of the forefinger to pinch a fine object. The movements of the fingertips are restricted in the main  $x$ – $y$  plane, which maps the flexion–extension plane of the forefinger. The workspace represents the reachable areas in which the manipulated object can be picked up. The magnitude of dexterity at each local point of the workspace is denoted by colours from blue to red.

functions [15]. For example, the longest digit of an anthropoid hand is the middle finger, whereas that of most prosimian hands is the ring finger.

### (b) Kinematic model

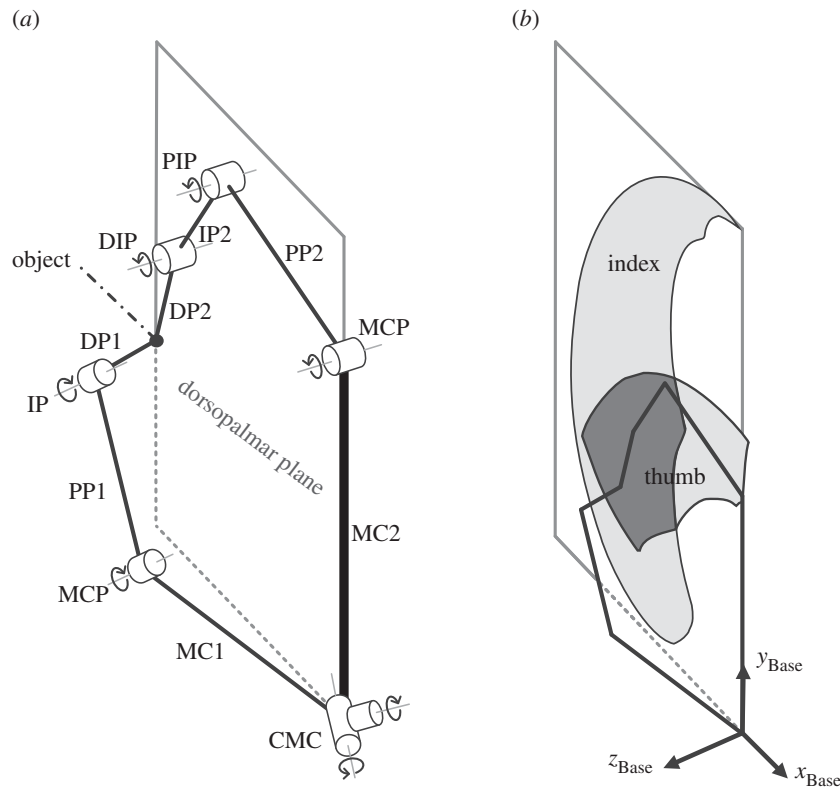
The thumb and forefinger are each simplified as a linkage model that imitates the corresponding hand skeletons (figure 2; see the electronic supplementary material, Note S2 for details of the kinematic model). The tips of the thumb and forefinger can pick up and hold a fine object such as a needle or a matchstick. The motion of the tip-pinch is assumed to be limited in the dorsopalmar plane of the hand (namely, the flexion–extension plane of the forefinger) [16]. The roots of the thumb and forefinger are postulated to be located at the same point, and the conjunct position is set as the origin of the base coordinate system (figure 2b). The  $y$ -axis of the coordinate system is aligned with the metacarpal of the forefinger, and the  $x$ -axis is directed outward from the palm. The locations of the tips of the thumb and forefinger in the base coordinate system are calculated from the joint angle data and the segment lengths using forward kinematics. For each digit, the position of its tip within the dorsopalmar plane  $\mathbf{x}_i = [x_i \ y_i]^T$  can be expressed as follows:

$$\mathbf{x}_i = H_i(\mathbf{q}_i), \quad (2.1)$$

where  $i = 1, 2$  separately represents the thumb and forefinger,  $\mathbf{q}_i$  denotes the joint angle vector of the respective digit, and  $H_i$  represents the kinematics equation (see the electronic supplementary material, Note S3). Because the hand sizes of different anthropoids vary greatly, the segment lengths of each individual are normalized to facilitate the comparison of diverse anthropoid hands. Every segment length is standardized relative to the overall length of the thumb and forefinger, and the whole length is represented by a constant of 1:

$$L_{MC1} + L_{PP1} + L_{DP1} + L_{MC2} + L_{PP2} + L_{IP2} + L_{DP2} = 1. \quad (2.2)$$

The ranges of motion of the joints refer to those of humans. The maximum flexion angles at the metacarpophalangeal



**Figure 2.** Kinematic model of the thumb–forefinger tip-pinch. (a) Simplified model of skeletons and joints. The model has seven skeletal segments, including the metacarpal (MC1), proximal phalanx (PP1), and distal phalanx (DP1) for the thumb and MC2, PP2, intermediate phalanx (IP2), and DP2 for the forefinger. The forefinger is modelled as a linkage that is connected by three hinge joints (metacarpophalangeal (MCP), proximal inter-phalangeal (PIP), and distal inter-phalangeal (DIP)) with their directions of rotation perpendicular to the flexion–extension plane of the forefinger (or called the dorsopalmar plane of hand). The thumb is modelled as a linkage that is connected by a universal joint (carpometacarpal (CMC)) and two hinge joints (MCP and inter-phalangeal (IP)). (b) Reachable workspace of the tip-pinch. The fingertip of the thumb touches the tip of the forefinger to pick up and manipulate a small object. The manipulated object can be moved in the overlapping region between the reachable areas of the fingertips of the thumb and forefinger within the dorsopalmar plane.

(MCP), proximal inter-phalangeal (PIP), and distal inter-phalangeal (DIP) joints of the forefinger are  $85.5^\circ$ ,  $102^\circ$ , and  $72^\circ$ , respectively [17]. The angular excursions in the carpometacarpal (CMC) joint of the thumb reaches  $37.6^\circ$  and  $46.3^\circ$  for flexion and extension, and abduction and adduction, respectively [18]. The maximum flexion values are  $50^\circ$  and  $80^\circ$  at the MCP and IP joints the thumb, respectively [19]. Using the normalized segment lengths and the ranges of motion of the joints, the reachable workspace of the tips of the thumb and forefinger within the dorsopalmar plane can be calculated (figure 2b; see the electronic supplementary material, Note S5 for the details of the numerical workspace calculation).

### (c) Measurement of manipulability

A commonly used measure of manipulative capability in robotics is the manipulability ellipsoid, which was proposed by Yoshikawa [20]. We refer to previous works [21,22] and derive the velocity manipulability ellipsoid for the in-hand manipulation between the tips of the thumb and forefinger for a given hand posture (see the electronic supplementary material, Note S4 for details of the derivation). The manipulability ellipsoid provides an intuitive perception of the ability of the fingertip to change position in different directions. The fingertip can move easily in the direction of the major axis of the ellipsoid, whereas movement in the direction of the minor axis is difficult. The motion capacity along the worse directions may become extremely difficult in certain postures. The fingertip would lose the ability to change position in the direction of minor axis of the manipulability ellipsoid for the specific posture, which is also called a singular state. However, the dexterous manipulation of the hand requires the capacity to quickly change the position

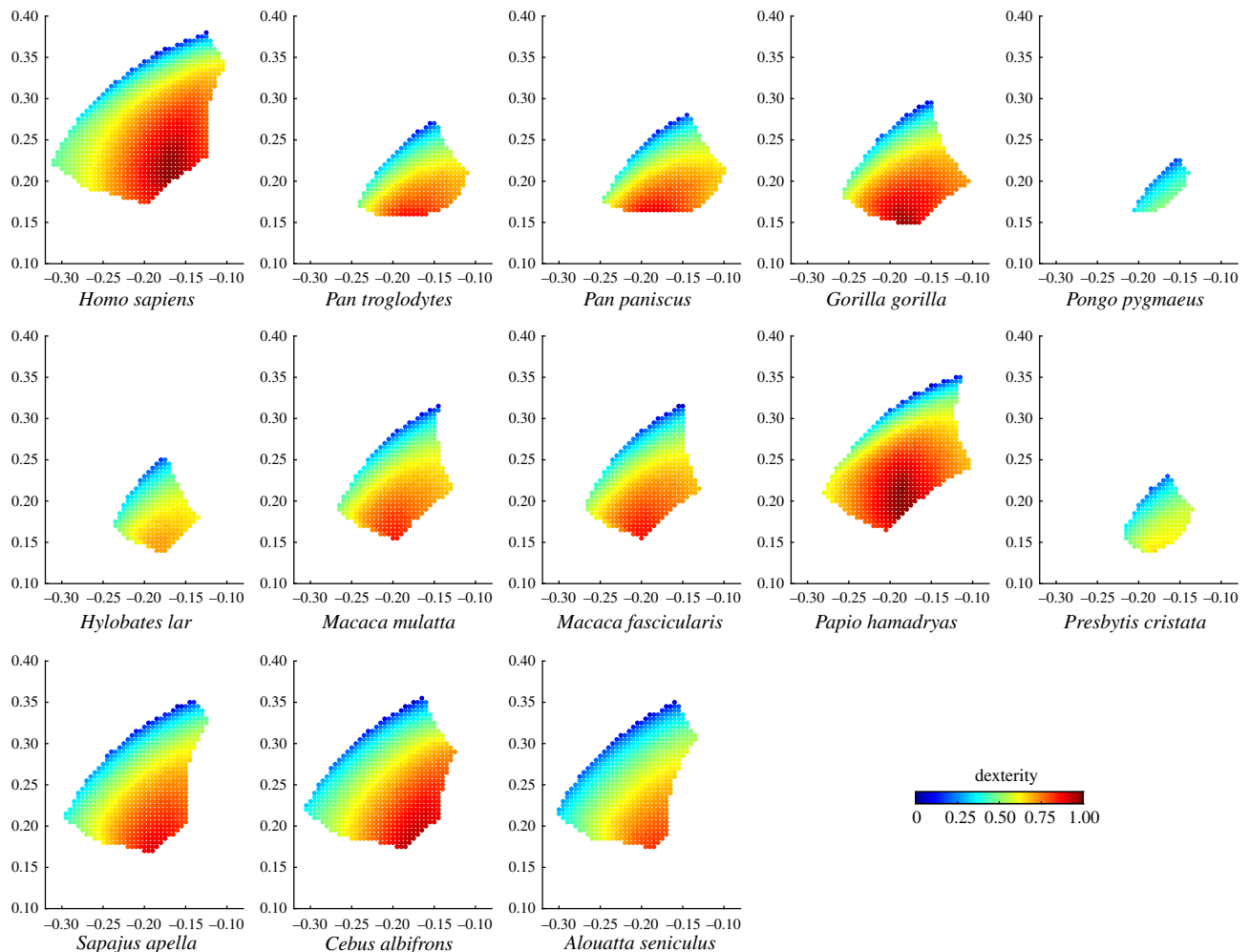
of the object over all directions between the thumb and forefinger. Thus, a singular posture is not desirable for dexterous manipulation and should be avoided as much as possible. A good measure of dexterous manipulative ability is the reciprocal of the condition number [23], which is determined by the following function:

$$\text{dex} = \frac{1}{\kappa} = \frac{\sigma_{\min}}{\sigma_{\max}}, \quad (2.3)$$

where  $\sigma_{\min}$  and  $\sigma_{\max}$  represent the minimum and maximum lengths of the axes of the velocity manipulability ellipsoid, respectively, and  $\kappa$  denotes the condition number introduced by Salisbury & Craig [24]. If the value of dexterity (dex) in a certain posture is one, then the fingertip has dexterous manipulative ability along any direction. However, if the value is near zero, then the fingertip can only move along some specific direction, and the manipulative ability is poor in this posture. The index of dex represents a measurement of dexterity in a particular configuration. To obtain a measure of the global behaviour of manipulative ability over the entire workspace of the fingertip, the GMI is proposed as follows:

$$\text{GMI} = \int_w \text{dex} \, dw, \quad (2.4)$$

where dex is the measure of the dexterous manipulative ability at one point of the fingertip's workspace  $w$  (see the electronic supplementary material, Note S5 for the numerical calculation of the GMI).



**Figure 3.** Manipulative potentials of anthropoid species assessed from hand proportions. The lengths of the hand segments are standardized for comparison among all anthropoid species. The mean values of the relative segment lengths are used to predict the manipulative potential for each species. A result with a large area of workspace and a bulk of dexterous workspace suggests that the species has better manipulative potential.

#### (d) Phylomorphospace

The phylomorphospace approach allows one to visualize the history of a clade's morphological diversification and infer the magnitude and direction of the morphological changes along any branch of the phylogeny [25]. This approach is used to inspect the evolutionary history of anthropoid hands by projecting the phylogenetic tree of anthropoids into the bivariate morphospace. The two-dimensional morphospace is constructed by two major variations of hand proportions among anthropoid species using a principal component analysis (electronic supplementary material, table S2). The morphological disparity between species can be intuitively distinguished from the distance of the PC scores in the morphospace. The mapping in the phylomorphospace approach is accomplished by reconstructing the hand proportions of hypothetical ancestors in the internal nodes using the maximum-likelihood method [26,27]. The phylomorphospace is calculated and drawn using the R package 'phytools' [28].

#### (e) Test of phylogenetic signals

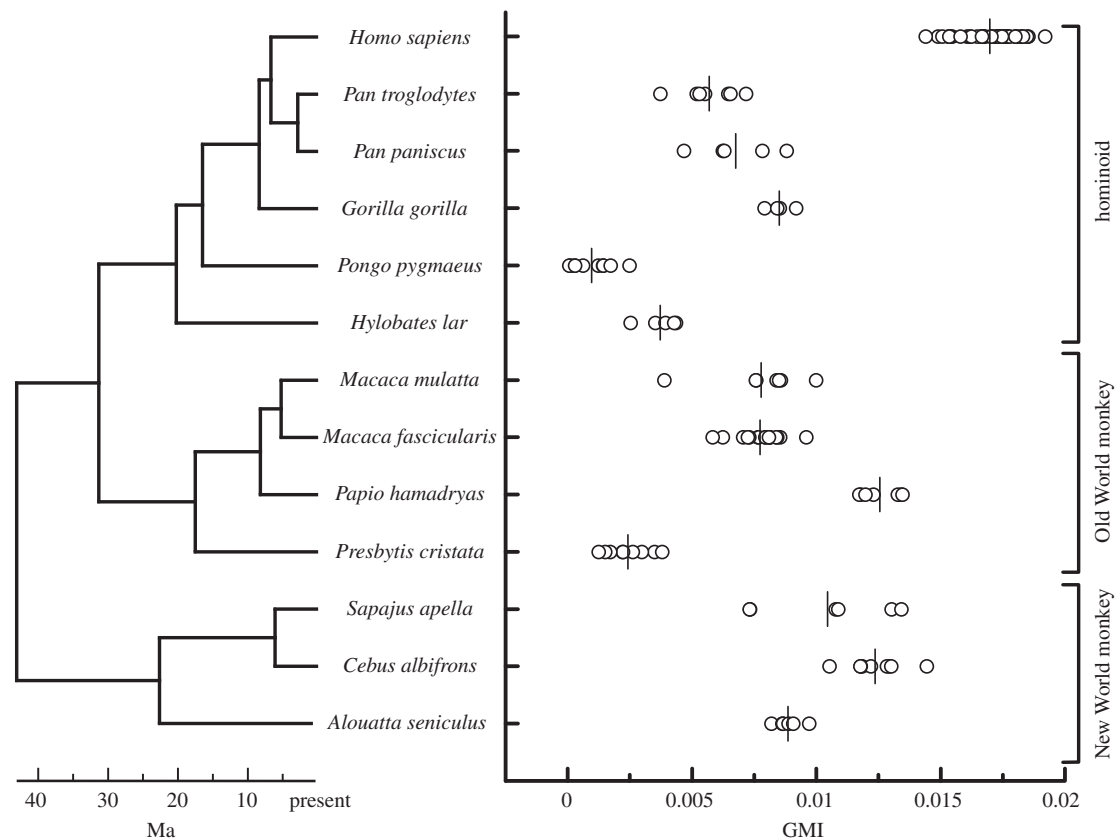
A phylogenetic signal indicates the tendency for phylogenetically related species to more closely resemble each other than more distant species [29,30]. In other words, closely related species should occupy the same portion in morphometric space. We implement a descriptive statistic,  $K$ , to quantify the strength of the phylogenetic signal in the hand proportions [30]. A high value implies a high degree of phylogenetic structure in the hand proportions. Hypothesis testing is then

conducted to determine whether the observed phylogenetic signal is statistically significant [29]. Statistical significance is set at 0.05. The phylogenetic signal is calculated and drawn in the R package 'phytools'.

### 3. Results

#### (a) Manipulative potentials of anthropoid hands with different segment proportions

We applied our functional model and estimated the manipulative potentials of anthropoid hands with different segment proportions, using the samples of monkeys, apes, and humans (see the electronic supplementary material, table S1 for details of the samples). Across all of the samples, the human hand has the largest workspace (figure 3; electronic supplementary material, table S1), and its size is significantly larger compared with other anthropoid hands ( $p < 0.001$ , Bonferroni-corrected post hoc tests in an analysis of variance (ANOVA); electronic supplementary material, table S3). The large workspace allows humans to pick up and conveniently move the manipulated object within the hand. In addition to the wide reachable workspace, another advantage of the human hand is its substantial area of dexterous workspace (figure 3). The large dexterous area permits the fingertip of the human hand to dexterously perform various complex



**Figure 4.** Phylogenetic relationship of anthropoids and corresponding manipulative potentials across all samples. The phylogenetic relationship of anthropoids is derived from the molecular genetic data [31]. Ma, million years ago; GMI, global manipulation index.

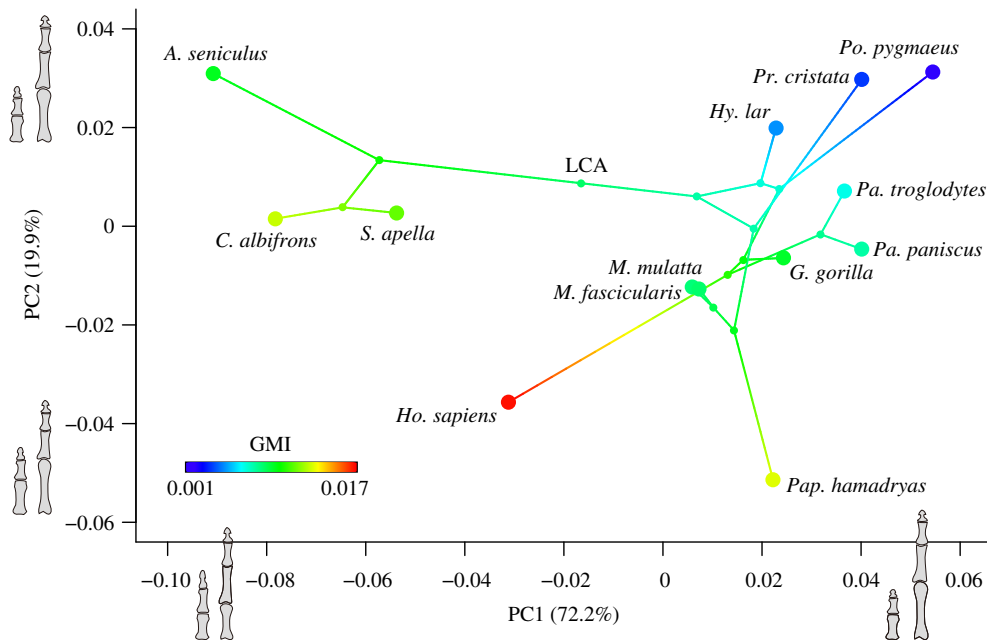
and fine manipulative tasks. GMI is developed to express the magnitude of the manipulative potential by considering both the workspace size and dexterity. According to the above definition, the hand proportions of humans possess the best manipulative potential among anthropoids. Although object manipulation skills contribute to survival [5], the species with the second best manipulative potential behind humans is not our close relative, the chimpanzee (*Pan troglodytes*). On the contrary, the hand proportions of chimpanzees provide them with poor potential of manipulation, from the aspects of both workspace size and dexterity. The manipulative potential of chimpanzees is significantly lower than that of gorillas and many monkeys (figure 4;  $p < 0.05$ , Bonferroni-corrected post hoc tests in an ANOVA; electronic supplementary material, table S4). The hand proportions of gorillas hold the best manipulative potential, whereas that of orangutans (*Pongo pygmaeus*) has the worst among the great apes. In the Old World monkeys, the hand of baboons (*Papio hamadryas*) has the best potential of manipulation and possesses a large dexterous workspace, which suggests that the hand proportions of baboons are suitable for manipulation. Moreover, it is unexpected that the manipulative potential in this species of Old World monkeys is distinctly better than that of the great apes (electronic supplementary material, table S4). It indicates that the hand proportions of baboons are more suited for manipulation than many extant hominoid hands. Within the New World monkeys, the hands of the white-fronted capuchins (*Cebus albifrons*) have better manipulative potential and a large workspace. The manipulative potentials of New World monkeys are also relatively better than those of great apes. Different species with different relative hand segment sizes can lead to great diversity of the potentials for manipulation, not only within species of Old

World monkeys, New World monkeys, and apes but also between them.

## (b) Effect of hand proportions on the manipulative potential

Anthropoid hands with different segment proportions present large disparities among their manipulative potentials. Thus, the question arises as to what changes of hand proportions cause the crucial diversity of manipulative potentials. A commonly used morphological feature is the length ratio between the thumb and fingers (the thumb proportion, or the opposability index in certain studies) [13,32–34]. The relative elongation of the thumb can facilitate the ability to oppose the thumb to the fingers and employ tip-to-tip precision grips when manipulating small objects. However, the thumb proportion is not an exact indicator to gauge or infer the manipulative potential. For instance, the manipulative potential of white-fronted capuchin monkeys (*Cebus albifrons*) is significantly lower than that of humans (figure 4;  $p < 0.001$ , Bonferroni-corrected post hoc tests in an ANOVA; electronic supplementary material, table S4) despite presenting a similar length ratio of the thumb (electronic supplementary material, figure S1). Different from the former studies based on the thumb proportion, our investigation further considers the detailed segment proportions in the thumb and forefinger.

Major trends in the variations of hand proportions among anthropoid species are investigated and summarized using a principal component analysis (electronic supplementary material, table S2; figure 5; see the electronic supplementary material, figure S2 for details on the variations between individuals). The primary changes of our anthropoid samples



**Figure 5.** Bivariate phylomorphospace plot of hand proportions mapped by manipulative potentials. The phylogeny presented in figure 4 is projected into the morphospace defined by the two principal components (electronic supplementary material, table S2) of the hand proportions among the anthropoid species mean values. Each tip node represents an extant species. The internal nodes of the hypothetical ancestral morphologies are constructed using the maximum-likelihood method. The magnitudes of the manipulative potentials based on the hand proportions are mapped onto the phylomorphospace using a colour gradient. The mapping is accomplished by estimating the manipulative potentials in the internal nodes using the constructed morphologies and interpolating the manipulative potentials of the ancestors along each branch.

along PC1 (accounting for 72.2% of the variance) are related to the shortening of the proximal and distal phalanges of the thumb and the lengthening of the metacarpal of the forefinger, whereas the primary changes along PC2 (accounting for 19.9% of the variance) are related to the shortening of the metacarpal of the thumb and the lengthening of the proximal and intermediate phalanges of the forefinger. Compared with humans, the thumb of white-fronted capuchin monkeys shows longer proximal and distal phalanges and a shorter metacarpal, and the forefinger shows longer proximal and intermediate phalanges and a shorter metacarpal (figure 5, along the negative direction of PC1 and the positive direction of PC2). Although the overall proportion of the thumb length in capuchin monkeys is similar to that of humans, the internal segment proportions in either the thumb or forefinger have large variations between them. The differences in the internal segment proportions in each digit are responsible for reducing the manipulative potential of the hands of capuchin monkeys relative to that of the hands of humans. This finding suggests that besides the overall proportion of the thumb, the internal segment proportions in each digit must be considered when inferring the manipulative potential of the hand.

The morphospace plot of hand proportions (figure 5) is overlapped by the manipulative potential to show the relationship between the morphological changes and manipulation. The results within the region of extant anthropoid hands demonstrate that the manipulative potential is enhanced along the negative directions of PC1 and PC2. The two directions reflect two variations of hand proportions that promote the manipulation. One is the lengthening of the proximal and distal phalanges of the thumb and the shortening of the metacarpal of the forefinger, and the other is the lengthening of the metacarpal of the thumb and the shortening of the proximal and intermediate phalanges of the forefinger. According to the results from our model, these

two features of morphological changes can be used to infer the manipulative potential. For example, a greater manipulative potential may be predicted when the hand proportions present morphological changes along the negative direction of both PC1 and PC2.

### (c) Evolution of hand proportions and manipulative potential

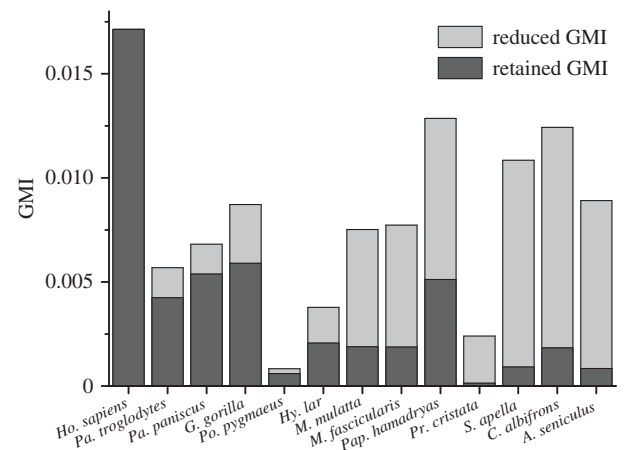
We implemented a phylomorphospace approach to inspect the evolutionary history of anthropoid hands and the corresponding manipulative potentials, although this method has some uncertainties of inferring ancestral states. A phylogenetic tree (figure 4) is mapped onto the morphospace of the hand proportions, and the major evolutionary changes can be visually tracked in the phylomorphospace (figure 5). The branch length and orientation of each clade represent the magnitude and direction of the changes in hand proportions along the branch. The evolutionary changes in hand proportions among all anthropoids conform to a phylogenetic structure ( $K = 1.16$ ,  $p < 0.01$ ; electronic supplementary material, figure S3), which indicates that relatives resemble each other and the hand proportions of the closely related species occupy the same portion of morphometric space. However, this evolutionary structure is not detected when only hominoids are considered ( $K = 0.67$ ,  $p = 0.424$ ; electronic supplementary material, figure S3). The lack of a phylogenetic structure implies that diverse shape changes occurred for hominoid hands along the branches in the phylomorphospace. After descending from a common ancestor, humans and chimpanzees underwent different changes of hand proportions along inverse directions, with humans showing changes along the negative directions of both PC1 and PC2 and chimpanzees showing changes along the positive directions of the two PCs (figure 5). It suggests that the human hand underwent the morphological changes

of lengthening all three segments of the thumb, which greatly promoted the manipulative potential, because the evolutionary process along the negative directions of both PC1 and PC2 represents the morphological changes that caused all three segments of the thumb to lengthen and the manipulative potential to rapidly increase. Conversely, the chimpanzee hand underwent the morphological changes of shortening all three segments of the thumb, which caused a marked decrease in manipulative potential (electronic supplementary material, figure S4). In addition, other apes, such as gibbons (*Hylobates lar*) and orangutans, also present shorter thumbs along the positive orientations of the two PCs during the evolutionary process. The branches of these great apes cross the branches of monkeys (electronic supplementary material, figure S4), and their potentials of manipulation are distinctly lower than those of most monkeys (figure 5).

In the evolutionary process of New World monkeys and certain Old World monkeys, the manipulative potentials improved slightly. The argument of manipulative potentials in New World monkeys is primarily based on the lengthening of the phalanges of the thumb and the shortening of the metacarpal of the forefinger (along the negative direction of PC1) after these monkeys descended from the last common ancestor (LCA) among the anthropoids. The enhanced manipulative potential in baboons (Old World monkeys) is mainly caused by the lengthening of the metacarpal of the thumb and shortening of the phalanges of the forefinger (along the negative direction of PC2), which occurred after they descended from the common ancestor with macaques. Although the manipulative potentials of these monkeys have increased, the degree of variation is not consistent with their large morphological changes as shown in the long lengths in the branches. The changes of hand proportions in these monkeys are along the direction of lower variation of manipulative potentials and not along the direction of rapid increases of manipulative potentials, such as in humans.

#### (d) Constraints of joint mobility on manipulative potential

If only the aspect of hand proportions is considered, then many monkeys indeed possess considerable potentials of manipulation. However, most monkeys cannot perform a multitude of complex manipulative tasks in daily life. One of the most important factors is the limited range of motion in the CMC joint of the thumb. The CMC joint is located at the base of the kinematic link of the thumb, and its constraints on joint mobility can greatly affect the opposability of the thumb to the forefinger. In the previous analysis of manipulative potential, we unify the ranges of motion of joints among all species by referring to those of humans, and specifically study the effect of variations in hand proportions on manipulation. In the following analysis, we will add the actual range of motion of each species measured by goniometry [18] to the functional model and reassess the manipulative potential from the two factors of hand proportions and joint mobility. Because the ranges of motion in the CMC joint of non-human anthropoids are smaller than that of humans, the manipulative potentials of the non-human anthropoid hands show a notable decrease (figure 6) when considering the factor of joint mobility. The reduction is reflected in a reduced size of the reachable workspace and a less dexterous workspace (electronic supplementary material, figure S5), which is



**Figure 6.** Manipulative potentials limited by the constraints of joint mobility. The whole portion of each column demonstrates the original assessed result of manipulative potential when considering a single factor of hand proportions, with unifying the range of motion of the joints among different anthropoid species by referring to those of humans. The dark grey columns illustrate the reassessed results of manipulative potentials when considering the two factors of both hand proportions and the ranges of motion in the joints. The range of motion for each species is derived from the actual measured data. The light grey columns indicate the reduction of manipulative potentials due to the limit of joint mobility for each species.

especially obvious for Old World monkeys and New World monkeys. The constraints of joint mobility in monkeys eliminate the better manipulative potentials provided by their specific hand proportions. This finding may provide an important explanation for the poor actual manipulative skills of monkeys whose hands are suited for manipulation from the aspect of hand proportions.

## 4. Discussion

In this study, we established a functional model to infer the manipulative potentials of extant anthropoid species from the morphology of hand proportions (see the electronic supplementary material, figure S6 for additional applications in relatively complete fossil hands). It should be noted that the assessed results from our model do not always reflect the observed manipulative abilities of the anthropoid species, because the actual manipulative ability relies on not only the morphology of hand proportions, but also many other factors. For example, many monkeys possess better potential manipulative abilities assessed by hand proportions, but they have poor manipulative skills in the real life. To explain the disparity, we add the factor of joint mobility and reassess the manipulative potential from the two factors of hand proportions and joint mobility. Our results show that the limited range of motion in the joints of the monkeys greatly limits the better manipulative potential given by their specific hand proportions. In addition, the lower intelligence of monkeys also limits the development of manipulative potential (see the electronic supplementary material, Note S6). These factors account for the disparity between the actual manipulative ability and the manipulative potential assessed from hand proportions.

In this paper, we specifically studied the effect of hand proportions on manipulation. Different from the rough morphological feature of a longer thumb [13,32–34], our results reveal two variations of hand proportions that facilitate

manipulative abilities: longer proximal and distal phalanges of the thumb and a shorter metacarpal of the forefinger; and a longer metacarpal of the thumb and shorter proximal and intermediate phalanges of the forefinger. The hand proportions of humans evolved along both of these variation schemes, which enabled the human hand to have the best manipulative potential among anthropoids. The best manipulative potential of the human hand also indicates that the design of a robotic hand for dexterous manipulation should imitate the proportions of the human hand.

In combination with the phylogenetically informed morphometric analyses of hand proportions, our results can also provide certain implications for hand evolution. The hand and brain are two important organs and are interdependent and coevolved in our anthropoid ancestors [7]. During the evolutionary process of anthropoids, the development of the brain is a distinct trait that greatly promotes manipulative abilities. Higher anthropoids are able to perform more complex manipulative tasks based on actual observed manipulative behaviours [2,35,36]. However, the hand has not coevolved with the brain to promote the manipulative skills in the non-human anthropoids (see the electronic supplementary material, Note S6 for an extended background). Although many animal species show an evolutionary convergence towards manipulative behaviour [37,38], the evolutionary changes of anthropoid hands may not have occurred to improve manipulative abilities from the aspect of hand proportions. For example, species that are phylogenetically close to humans, such as chimpanzees, who display high intelligence, present a decreased potential for manipulation, which occurred after their descent from a common ancestor with humans.

On the contrary, the changes of non-human anthropoid hands are much more affected by the selective pressure for locomotion (e.g. [15,33,39,40]). Based on the morphological changes of hand proportions, our results indicate that the poor manipulative potentials in the great apes are a consequence of their relatively long fingers and short thumbs. This morphological characteristic of ape hands is most probably driven by suspensory locomotor constraints. To adapt the suspensory behaviours, the fingers of the apes have to elongate so that to flex completely around the branches in a firm hook-like grasp, while a long thumb would hinder the locomotor way (see the electronic supplementary material, Note S7 for an extended explanation). Our results also show that the changes of hand proportions in certain New World and Old World monkeys can lead to the enhancement of manipulative potentials. However, such changes cannot be well explained by the selection pressure for manipulation because the hand proportions in these monkeys have notably changed, whereas the manipulative potentials have only been slightly enhanced (figure 5). The manipulative potentials were likely strengthened

only as a by-product of an adaption for locomotion in these monkeys. The enhancement for the manipulative potentials in the New World monkeys is mainly based on the lengthening of the phalanges of the thumb and the shortening of the metacarpal of the forefinger. These morphological changes are typically explained as an adaption for arboreal locomotion in a fine-branch milieu [41,42]. With respect to the other Old World monkeys, the changes in the hand proportions in baboons have led to a relative enhancement of manipulative potential by the proportional shortening of the phalanges of the finger and the lengthening of the metacarpal of the thumb. The morphological changes most probably occurred as a response to the stresses induced by terrestrial digitigrade locomotion [33,43]. Therefore, according to the explanations from the existing literature, the morphology of hand proportions in non-human anthropoid hands may not be a compromise between the needs for manipulation and locomotion, but more likely evolve for locomotion.

In addition to the hand proportions, the locomotor behaviours also restrict manipulation by limiting the range of motion in the joints. Joint mobility can be regarded as a mirror of the patterns of locomotor behaviours (see the electronic supplementary material, Note S8 for an extended interpretation). The joints of non-human anthropoid hands are subjected to vast compressive loads under the locomotor behaviours. For preventing possible damage like dislocation under the large compressive loads, the ligaments in the joints become tight to increase the stability, which costs the range of motion (see the electronic supplementary material, Note S8 for a detailed interpretation). This additional limit on the range of motion enormously constrains the manipulative potentials, especially for monkeys. The limit of the range of motion in the joints of monkeys eliminates the better manipulative potentials of monkeys given by their hand proportions. Thus, from the aspect of joint mobility, the limit of range of motion in non-human anthropoid hands costs manipulation for the sake of locomotion. It also supports the viewpoint that non-human anthropoid hands have more likely evolved under selective pressure for locomotion than manipulation.

**Data accessibility.** All data can be found in the electronic supplementary material.

**Authors' contributions.** M.L. devised the study, analysed the data, and drafted the manuscript. C.X. directed the research and wrote the manuscript. D.H. created the model and participated in the analysis. All authors gave final approval for publication.

**Competing interests.** The authors declare no conflicts of interest.

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